

TO THE EDITOR:

Identifying *STRN3-RARA* as a new fusion gene for acute promyelocytic leukemia

Qi Zhang,^{1,*} He Li,^{2,*} Xuelan Chen,^{1,*} Fan Gu,¹ Lanxin Zhang,¹ Lu Zhang,¹ Tong Chen,³ Qiang Chen,⁴ Wentong Meng,⁵ Yu Wu,² Hong Chang,² Ting Liu,² Chong Chen,¹ Hongbing Ma,² and Yu Liu¹

¹Department of Hematology and Institute of Hematology, State Key Laboratory of Biotherapy and Cancer Center, and ²Department of Hematology, Institute of Hematology, West China Hospital, Sichuan University, Chengdu, Sichuan, China; ³Sichuan Hua Xi Kindstar Medical Diagnostic Centre, Chengdu, Sichuan, China; ⁴Sichuan Neo-Life Stem Cell Biotech Inc, Institute of Blood Transfusion, Chinese Academy of Medical Sciences and Peking Union Medical College, Chengdu, Sichuan, China; and ⁵Laboratory of Stem Cell Biology, State Key Laboratory of Biotherapy, West China Hospital, Sichuan University, Chengdu, China

Here we report a new fusion gene, *STRN3-RARA*, in acute promyelocytic leukemia (APL). It cooperates with *UTX* deficiency to drive full-blown APL in mice. Although *STRN3-RARA* leukemia quickly relapses after all-trans retinoic acid treatment, it can be restrained by cepharanthine.

Acute promyelocytic leukemia (APL) is a distinct subtype of acute myeloid leukemia that is generally driven by *PML-RARA*, resulting from a balanced chromosomal translocation t(15;17) (q24.1;q21.2).^{1,2} However, approximately 2% of patients with APL carry other variants of *RARA* fusions, which pose challenges for diagnosis and treatment.^{3,4} Here, we report a novel t(14;17) translocation in a patient with APL that gave rise to a new fusion gene, *STRN3-RARA*. *STRN3* forms the striatin-interacting phosphatase and kinase complex, which couples kinases to protein phosphatase 2A and regulates their activities.⁵ Like other patients with APL with *RARA* variants, this patient quickly relapsed after the initial response to all-trans retinoic acid (ATRA) treatment.^{6,7} Mechanistically, we demonstrated that *STRN3-RARA*, in cooperation with *UTX* deficiency (which was mutated in the patient), drove APL formation in mice. Additionally, we found that *STRN3-RARA* cells were specifically susceptible to the tumor necrosis factor α (TNF α) pathway. Inhibition of this pathway by cepharanthine, a Food and Drug Administration–approved drug, effectively suppressed the growth of U937 cells with *STRN3-RARA* expression, as well as relapsed APL cells from the patient. Taken together, our study identified a new APL fusion gene, *STRN3-RARA*, and validated it as a functional APL driver, providing insight into the diagnosis and treatment of patients with nonclassic APL.

Informed consent was obtained from the patient's family. All mouse experiments were approved by the Institutional Animal Care and Use Committee of West China Hospital of Sichuan University.

A 50-year-old man (WCH01 [patient 01 with *STRN3-RARA* fusion in West China Hospital]) was admitted to the hospital with leukopenia and coagulopathy. In the bone marrow (BM), 91.5% of cells were hypergranular promyelocytes (Figure 1A), combined with immunophenotype and other clinicopathological features; this WCH01 patient was diagnosed with APL (Figure 1B and supplemental Table 1, available on the *Blood* website). Further transcriptomic analysis of WCH01 showed enrichment of gene signatures of APL_PML_RARA⁸ (Figure 1C).

However, the classic chromosome translocation t(15;17) was not detected using the fluorescence in situ hybridization test with *PML* and *RARA* probes (supplemental Figure 1A). Instead, a new translocation between chromosomes 14 and 17 was found in the karyotyping result, which was further confirmed by RNA sequencing (RNA-seq) and Sanger sequencing (Figure 1D-F). This fusion happened between *STRN3* exon 3 and *RARA* exon 3, resulting in a new *STRN3-RARA* fusion. Additionally, several gene mutations, including *UTX* frameshift mutation, were detected by the RNA-seq of WCH01 APL cells (Figure 1G and supplemental Figure 1B).

Initially, the WCH01 patient achieved complete remission with standard ATRA and arsenic trioxide (ATO) combination treatment, followed by consolidation with anthracycline-based therapy and maintenance with ATRA and arsenic trioxide. But after 10 months, he quickly relapsed and became resistant to ATRA and ATO treatment (Figure 1H and supplemental Figure 1C-E). A novel *RARA* missense mutation (S232F) was detected in the relapse sample (supplemental Figure 1F). The WCH01 patient then received a combination of ATRA and venetoclax. But unfortunately, no response was seen. He quickly developed severe pneumonia and eventually died from respiratory failure after 14 days of treatment with ATRA and venetoclax (Figure 1H).

To investigate the role of the *STRN3-RARA* (SR) fusion, we cloned Flag-tagged SR complementary DNA into a green fluorescent protein (GFP)-expressing retrovirus-based construct (pMSCV-Flag-SR-IRES-GFP) and transduced it into cell lines. Like *PML-RARA* and other reported *RARA* variant fusion proteins, the SR protein was predominantly located in the nucleus and was sensitive to ATRA treatment^{9,10} (Figure 2A-B and supplemental Figure 2A). Transcriptomics analysis of U937 cells with SR showed that APL_PML_RARA-upregulated genes were significantly positively enriched in SR cells (Figure 2C), which was consistent with the significant overlap of the differentially expressed genes in *PML-RARA* and those in *STRN3-RARA* (supplemental Figure 2B-C). We further conducted CUT&Tag sequencing of Flag-SR U937 cells to discover the SR-binding

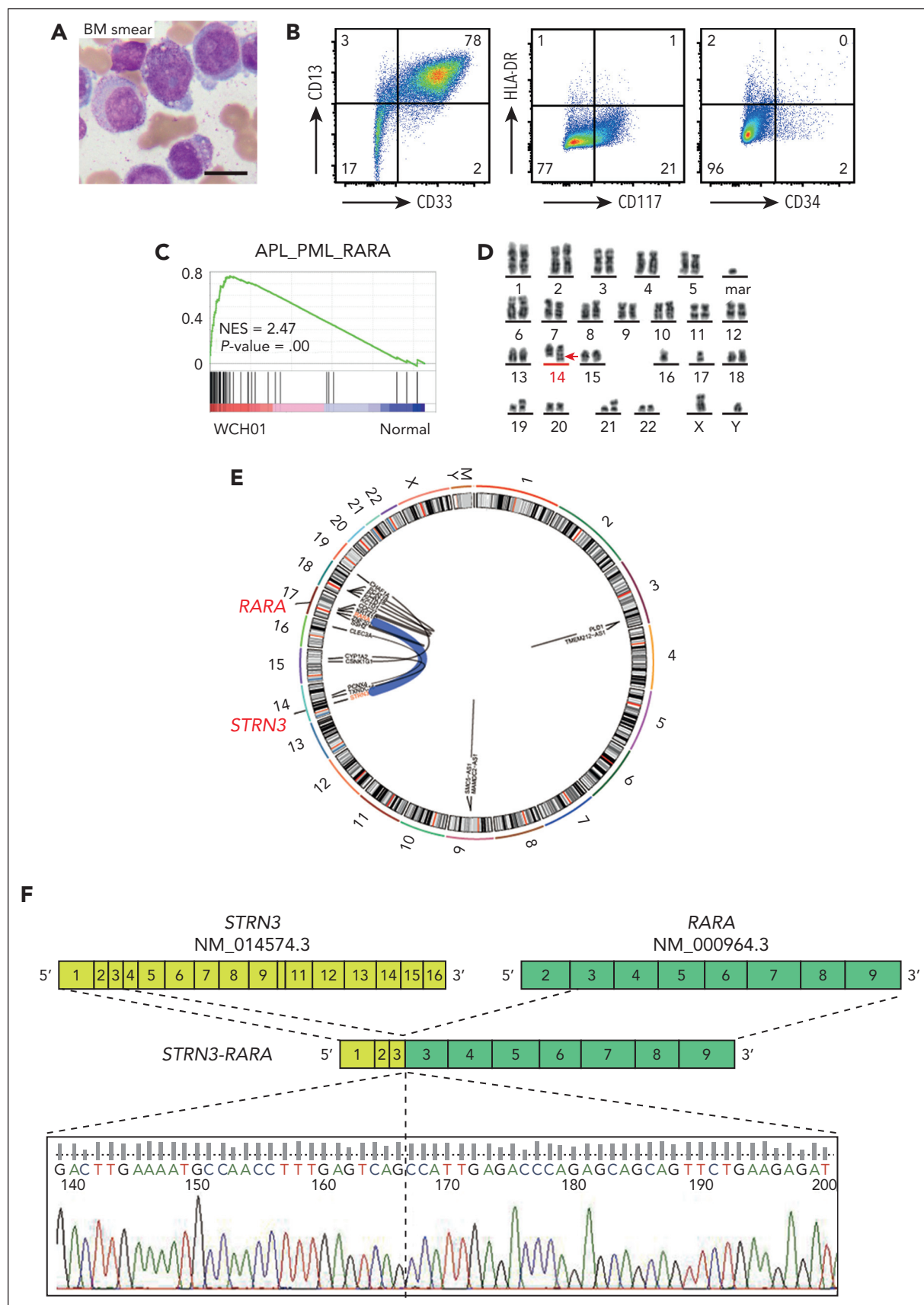


Figure 1. Identifying an APL with *STRN3*-*RARA* translocation. (A) BM smear of the WCH01 patient before treatment. Scale bar, 10 μ m. (B) Flow cytometry analysis of the WCH01 patient's BM cells before treatment. The expression of CD13, CD33, CD117, CD34, and HLA-DR is shown. The numbers indicate the percentage of different cell

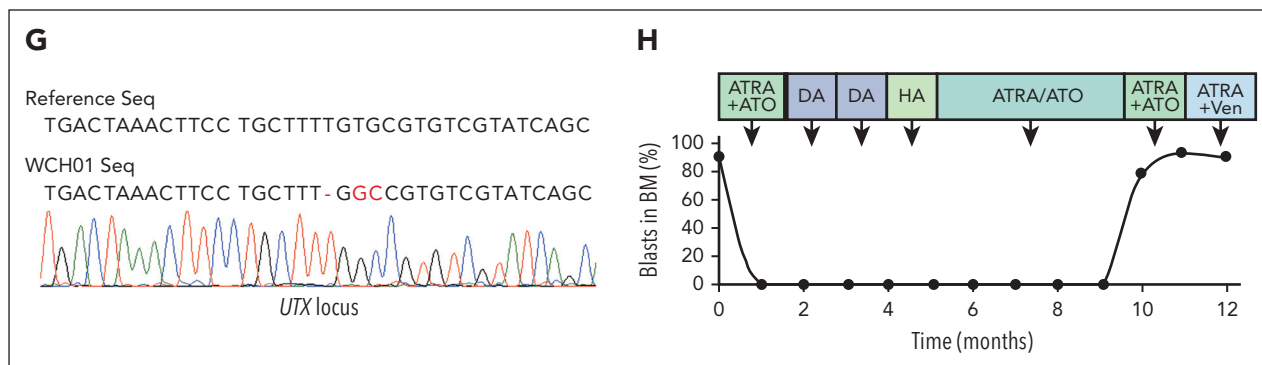


Figure 1 (continued) populations. (C) Gene Set Enrichment Analysis (GSEA) showing the positive enrichment of the APL_PML_RARA gene set in the WCH01 patient, compared with normal gene sets (NES = 2.47; $P = .00$). (D) Cytogenetics analysis of the WCH01 patient's BM cells before treatment. The karyotype analysis of the WCH01 patient reveals the following abnormalities: 45, XY, add(6)(q13), der(14)t(14,17)(q12;q21) dup(17)(q21q25), -16, -17, add(18)(p11), der(19)t(15;19)(q22;p13.3), add(21)(p11), +mar, inc[20]. The red arrow highlights the t(14,17) chromosome translocation. (E) The circular plot showing an overview of the top 10 fusion events between locations in chromosomes. (F) Schematic representation and Sanger sequencing of the *STRN3*-*RARA* fusion. The yellow blocks represent exons of the *STRN3* gene (NM_014574.3), and the green blocks represent exons of the *RARA* gene (NM_000964.3). Numbers in the blocks indicate the exon number. The Sanger sequencing at the breakpoint site is shown at the bottom. (G) Sanger sequencing of the *UTX* locus obtained from the polymerase chain reaction product of the WCH01 patient. (H) Percentage of blast cells in the WCH01 patient's bone marrow during treatment. The blocks indicate the drug treatments administered during the period. DA, daunorubicin and cytarabine; NES, normalized enrichment score; Ven, venetoclax.

sites (Figure 2D) and found that these SR-binding sites significantly overlapped with PML-RARA-binding genes¹¹ (Figure 2E). Combined with RNA-seq data, we found that SR protein directly bound to genes and upregulated many APL-related genes or downregulated multiple differentiation-related genes (supplemental Figure 2B-E). These results suggest that, like PML-RARA, the *STRN3*-*RARA* oncogenic protein exerts both repressive and activating functions through direct binding.¹¹

To investigate the role of SR in APL genesis, we transduced the hematopoietic stem and progenitor cells with SR and followed by transplantation into sublethal irradiated recipient mice. We found that SR alone was insufficient to induce a full-blown APL in mice over a 5-month period, which is consistent with the previous observations in the *PML*-*RARA* mouse model¹² (supplemental Figure 3). Considering that the frameshift mutation of a known tumor suppressor gene *UTX* was detected¹³⁻¹⁶ (Figure 1G), we investigated the role of SR in murine APL formation with *Utx* single guide RNA (*sgUtx*). The results showed that all *sgUtx*;SR cohort mice had developed leukemia by 175 days after transplantation and none of the *sgUtx*;Vector control mice were sick (Figure 2F and supplemental Figure 4A-H). The peripheral blood smear results displayed blast cells with abnormal promyelocytic morphology (Figure 2G). Pathological examinations of the BM, spleen, and liver also revealed the infiltration of leukemic cells (supplemental Figure 4I-K). Besides, recipient mice transplanted with *sgUtx*;SR APL cells received ATRA or vehicle treatment, resulting in a reduction of GFP + mCherry + APL cells in the peripheral blood. The induction of differentiation was observed through blood smear and RNA-seq analysis (Figure 2H-I and supplemental Figure 5). Furthermore, the treatment significantly extended the survival of the recipient mice (Figure 2J). These results indicate that SR fusion with *Utx* deficiency drives APL.

To elucidate the molecular mechanisms underlying the cooperation between SR fusion and *UTX* loss, we studied the gene expression upregulated or downregulated in the WCH01 compared with normal human BM cells or in U937 cells with

sgUTX;SR compared with that with empty vector and found that 454 upregulated genes (common_up_454) and 587 downregulated genes (common_down_587) were significantly overlapped^{17,18} (supplemental Figure 6A-C). According to these gene expression patterns, we divided the common_up_454 genes (or common_down_587) into 3 groups and labeled them with different colors. Genes in the green group were likely regulated mainly by SR, genes in the orange group were regulated mainly by *UTX*, and genes in the blue group seem to be under the control of both *UTX* and SR (supplemental Figure 6D-E).

To find a new inhibitor for *sgUTX*;SR, we searched for candidate drug-targetable genes that were specifically upregulated by *sgUTX*;SR but not *PML*-*RARA*. TNF gene and pathway were at the top of the list (Figure 2K-L). Therefore, we examined the impact of TNF α inhibitor ceftaraphine, a Food and Drug Administration-approved drug, on the growth of U937 cells with vector, *sgUTX*, SR, or *sgUTX*;SR.¹⁹ Our results demonstrated that U937 cells with SR or *sgUTX*;SR were more sensitive to ceftaraphine than control cells (Figure 2M). Furthermore, although clinical SR APL cells collected from WCH01 patient who relapsed showed no response to ATRA treatment, their viability was reduced by ceftaraphine treatment (Figure 2N). These results suggest that ceftaraphine could serve as a potential therapeutic method for patients with APL with *STRN3*-*RARA* fusion.

In conclusion, we have identified a novel fusion gene, *STRN3*-*RARA*, in a patient with APL. This SR APL displayed similar clinical and pathological characteristics as patients with classical APL carrying *PML*-*RARA*. Additionally, we have demonstrated that SR, with the *Utx* loss, drove APL genesis in mice, revealing the oncogenic role of this fusion gene in leukemogenesis. Furthermore, we found that SR APL cells collected from the WCH01 patient who relapsed were sensitive to the inhibitor of TNF α signaling pathway. Taken together, our findings not only advance the understanding of APL disease but also provide valuable insights for the diagnosis and treatment of patients with APL variants.

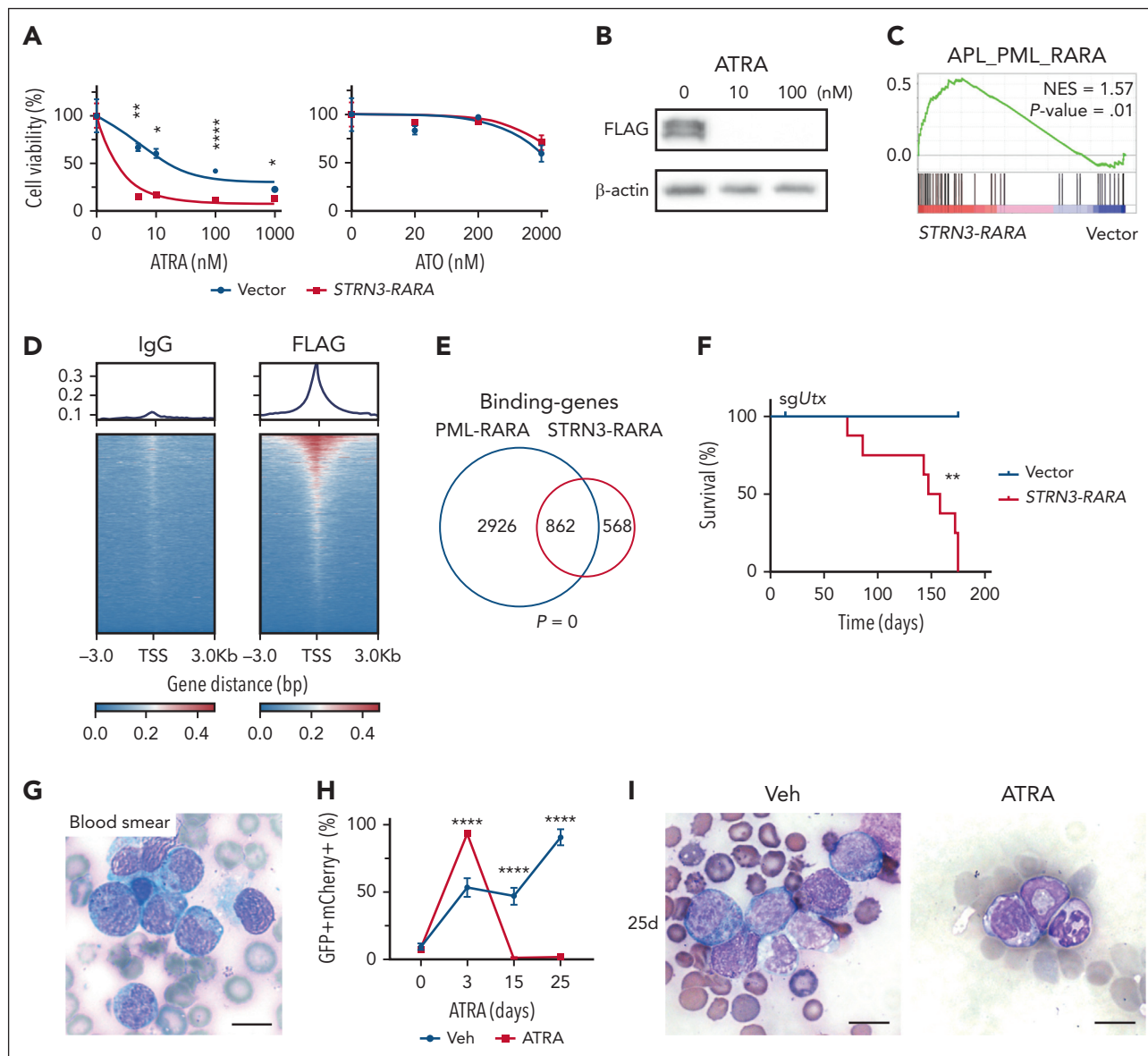


Figure 2. The oncogenic role of *STRN3-RARA* fusion gene in leukemogenesis. (A) Treatment curve of U937 cells with ATRA and ATO. Cells were transduced with vector or *STRN3-RARA*. Means $\pm$ standard deviation (SD), $n = 3$; $*P < .05$, $**P < .01$, $****P < .0001$ (2-way analysis of variance [ANOVA]). (B) Western blot analysis of ATRA-treated 293T cells harvested 2 days after treatment. 293T cells were transduced with *Flag-STRN3-RARA*. (C) GSEA showing positive enrichment of the *APL_PML_RARA* gene set in U937 cell with *STRN3-RARA* overexpression, compared with vector (NES = 1.57; $P = .01$). (D) Levels of *STRN3-RARA* bound at the transcription start site (TSS) in U937 cells transduced with *STRN3-RARA*, measured by the CUT&Tag analysis. (E) The Venn diagram showing the overlap of binding-genes in *STRN3-RARA* and *PML-RARA* (hypergeometric test). (F) Kaplan-Meier tumor-free survival curves of recipient mice in vector and *STRN3-RARA* group with *Utx* knockout, $n = 6$ (vector), $n = 8$ (*STRN3-RARA*); $**P < .01$ (log-rank test). (G) Blood smear of *sgUtx;SR* recipient mouse before sacrificed. Scale bar, 10 μ m. (H) The percentage of *sgUtx;SR* cells (GFP + mCherry + cells) along with vehicle and ATRA treatment. The x-axis indicates the number of days after treatment. Means $\pm$ SD, vehicle, $n = 5, 4, 4, 2$; ATRA, $n = 5, 4, 4, 4$; $****P < .0001$ (2-way ANOVA). (I) Blood smears from *sgUtx;SR* mice following a 25-day treatment with vehicle or ATRA. Scale bar, 10 μ m. (J) Survival curve of *sgUtx;SR* mice following a 25-day ATRA or vehicle treatment. $n = 5$; $*P < .05$ (log-rank test). (K) Heat map showing the 140 drug targetable genes that were specifically upregulated in U937 cells with *sgUtx;SR* compared with that with empty vectors and not overlapped with the upregulated genes due to *PML-RARA*. (L) GSEA showing the positive enrichment of the *HALLMARK_TNFA_SIGNALING_VIA_NFKB* gene set in *sgUtx;SR* U937 cells, compared with vector (NES = 1.67; $P = .00$). (M) Relative viability of U937 cells treated with 1 μ M cepharanthine. Vector, *sgUTX*, *SR*, and *sgUTX;SR* groups are shown. Means $\pm$ SD, $n = 3$; $*P < .05$, $**P < .01$ (1-way ANOVA). (N) Drug treatment survival curve of WCH01 relapsed cells. ATRA and cepharanthine were treated respectively. Means $\pm$ SD, $n = 3$; $**P < .01$, $****P < .001$ (2-way ANOVA). IgG, immunoglobulin G; Veh, vehicle.

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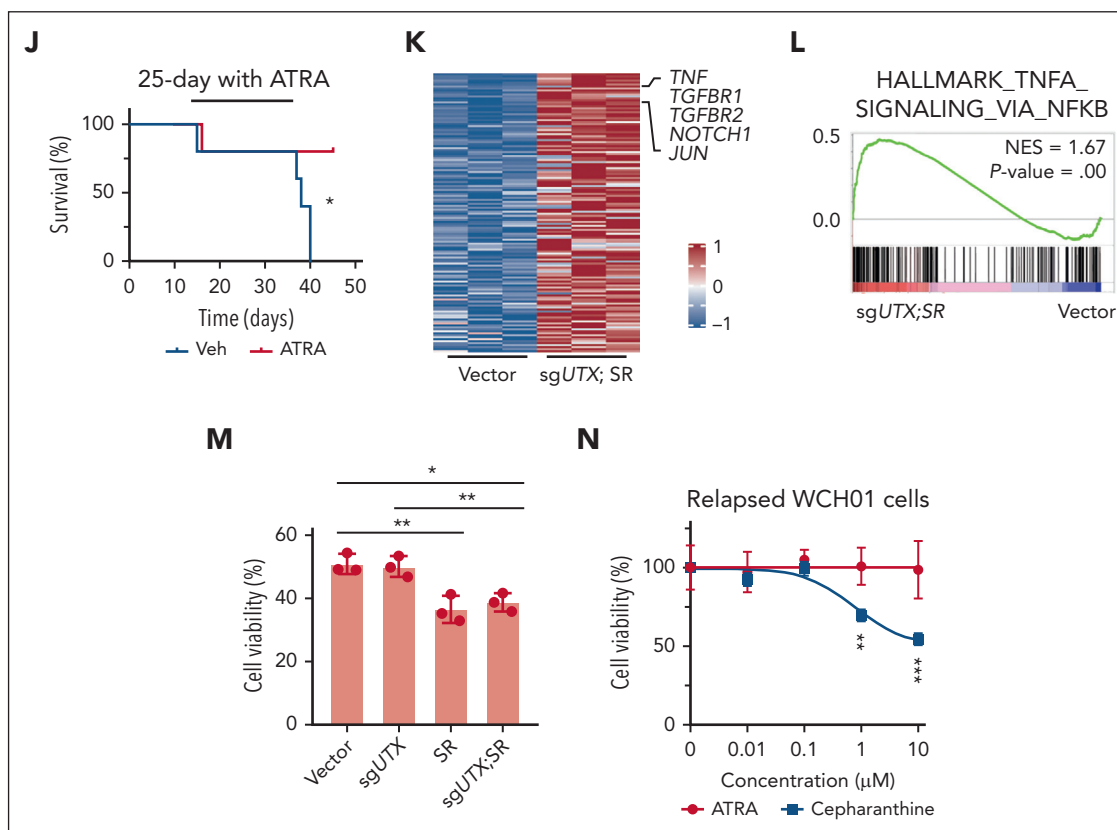


Figure 2 (continued)

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Authorship

Contribution: Y.L. and H.M. designed and supervised this study; T.L., C.C., and W.M. were involved in the study design; H.M., H.L., Y.W., and H.C. recruited and treated the patient; Q.Z., F.G., Lanxin Zhang, and T.C. designed and performed experiments; X.C. and Lu Zhang performed the bioinformatic analyses; Q.C. provided human cells; and Y.L., H.M., C.C., Q.Z., and X.C. analyzed the data and wrote the manuscript.

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ORCID profiles: X.C., 0000-0003-3987-4971; Lu Zhang, 0000-0001-9337-093X; W.M., 0000-0002-9592-5668; Y.W., 0000-0001-8708-9711; T.L., 0000-0003-1816-5551; H.M., 0000-0002-1174-4078; Y.L., 0000-0002-1193-7830.

Correspondence: Hongbing Ma, Department of Hematology and Institute of Hematology, West China Hospital, Sichuan University. NO. 37 Guo Xue Xiang St, Chengdu 610041, China; email: hongbingma@foxmail.com; and Yu Liu, State Key Laboratory of Biotherapy and Cancer Center, West China Hospital, Sichuan University, 3-17 Renming Rd, Chengdu 610041, China; email: yuliuscu@scu.edu.cn.

Footnotes

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*Q.Z., H.L., and X.C. contributed equally to this work.

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Data are available upon request to the corresponding authors (hongbingma@foxmail.com and yuliuscu@scu.edu.cn).

The online version of this article contains a data supplement.

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